HONEYWELL EMBEDDED ENGINEERING TEAMS — PA · IA · BA

AI Champions Programme

Module 10

Agent Prism for Monitoring, Observability and Enterprise Control

The monitoring lens from Module 01, now hands-on: traces, failure patterns, token/cost leakage, quality signals and adoption — made visible, not assumed.

DURATION

2 Hours · Module 10 of 12

FORMAT

Guided Ingest + Compare-Runs Lab

DELIVERED FOR

Honeywell Embedded Engineering

NIIT StackRoute × RPS Programme Design

How We'll Spend These Two Hours

Six connected moves — from why monitoring matters to telemetry tied back to engineering KPIs.

01

Recall & Why Monitoring Matters

15 min

Agent Prism as the monitoring lens from Module 01 — now put into practice.

02

Traces, Replay & Drift

20 min

Stepping back through what an agent actually did, and whether output is drifting.

03

Failure Patterns & Cost Leakage

20 min

Common ways agent runs go wrong, and where token spend quietly disappears.

04

Quality Signals, Adoption & Thresholds

15 min

Turning raw telemetry into scorecards, alerts, and an operational review rhythm.

05

Ingest, Inspect, Compare

40 min

Load real run telemetry, inspect failures, and compare successful vs. poor-quality runs.

06

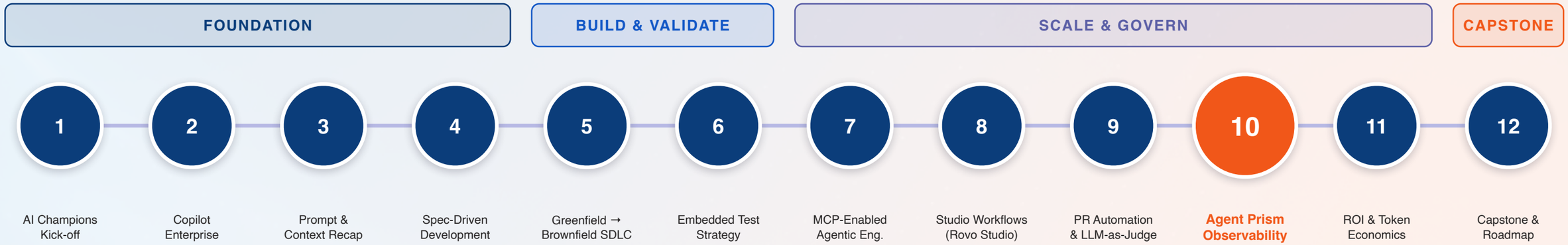
Connect to Engineering KPIs

10 min

Tie today's findings back to Module 06's test evidence and Module 09's PR quality gates.

Module 10 in the 12-Module Journey

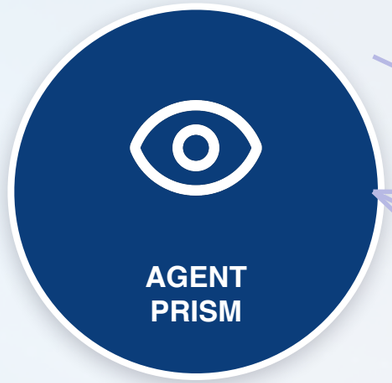
The concept from Module 01 becomes hands-on practice — right before it feeds the ROI model in Module 11.



Why this matters: Every module since Module 01 has generated something Agent Prism can observe — Copilot sessions, MCP workflows, PR quality gates. Module 11's token economics and ROI model are built entirely on what gets measured here.

Recall: Agent Prism, From Concept to Practice

Module 01 introduced this as the monitoring lens. Today, every spoke gets a hands-on exercise.



Traces & Replay

Today: step through a real agent run, action by action.

Slide 7

Failure Patterns

Today: categorize why runs fail, not just that they did.

Slide 8

Token Usage & Cost Leakage

Today: find exactly where spend disappears in a real run.

Slide 9

Quality & Control Signals

Today: turn signals into a scorecard you'd actually use.

Slide 10

Adoption Tracking

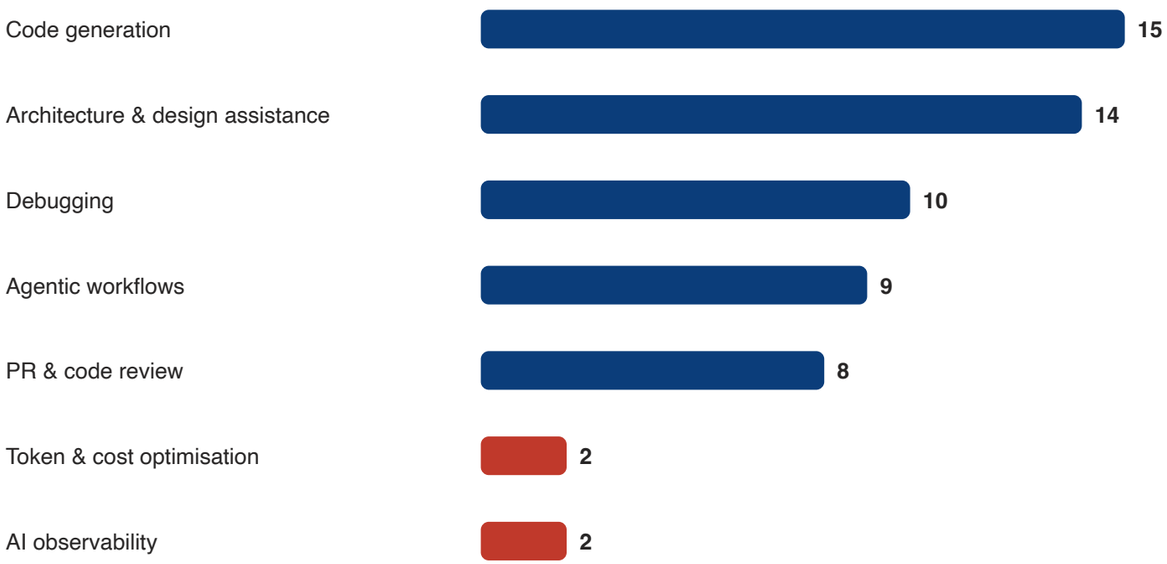
Today: see who's using AI-assisted workflows, and how often.

Slide 10

The Blind Spot This Cohort Didn't Ask For

Honest calibration data — the least-requested capability is arguably the one that makes every other one provable.

Which AI Capabilities Would Help Your Team Most?



(Full ranked list of 14 capabilities — top 5 and bottom 2 shown)

Tied for last: 2 of 19

AI observability and token/cost optimisation ranked dead last, tied with build & CI assistance — the same 2 of 19 who ranked everything else highly.

That's not surprising — nobody asks to monitor a tool before they trust it enough to use it daily. But 16 of 19 already use AI daily or regularly (Module 02). The trust is here. The visibility isn't yet.

Why Monitoring Agentic Engineering Matters

Every earlier module produces agent activity — without this one, most of it is invisible.

WITHOUT OBSERVABILITY

- An agent failure is invisible until someone notices bad output downstream.
- Token spend accumulates with no one able to say where it went, or why.
- “Is adoption actually working?” gets answered with anecdotes, not data.
- A quietly drifting agent keeps running the same way until it causes real damage.
- Governance conversations happen with no evidence to ground them in.

WITH AGENT PRISM

- A failed run is traceable to the exact step where it went wrong.
- Token spend is attributable to specific workflows — leakage gets named, not guessed at.
- Adoption tracking gives a real number: who's using what, and how often.
- Drift shows up as a trend before it becomes an incident.
- Governance & ROI conversations in Module 11 start from evidence, not opinion.

Observability Concepts: Traces, Replay & Drift

Three ways of asking the same question: what actually happened, and is it changing?

Traces

The full step-by-step record

Every tool call, decision & context pull an agent made, in order — the raw material everything else is built from.

Replay

Step back through a run, exactly

Re-run a trace step by step to see precisely where a good or bad outcome was decided — not just what the final output was.

Prompt / Output Drift

Change you'd otherwise miss

The same task type producing steadily different results over weeks — invisible run by run, obvious over a trend line.

Failure Patterns: What Goes Wrong, and How to Spot It

Six categories — naming the pattern is most of the work of fixing it.

Tool Call Failure

A build, test or API call fails mid-workflow — visible in the trace as a hard stop.

Retry Loops

The agent repeats the same failing action without adapting — a clear trace signature.

Context Misses

The agent acts without context it should have had — visible as an unexplained wrong turn.

Output Drift

Quality degrades gradually across similar tasks — only visible in aggregate, not per-run.

Scope Creep

An agent modifies more than the task required — the Module 07 boundary being tested.

Silent Low Confidence

A run completes but should have escalated — the Module 09 judge miscalibrated.

Today's lab exercise: you'll be given real run telemetry and asked to identify which of these six patterns explains what happened — before you're told the answer.

Token Usage & Cost-Leak Visibility

Module 03 taught you to spend tokens deliberately. This is where you find out if it worked.

Where Spend Is Expected

Task framing, scoped code context, targeted test generation — the Module 03 discipline, working as intended.

Normal

Where Spend Gets Suspicious

Repeated re-explanation of the same context, oversized repository dumps, retry loops re-sending full history.

Investigate

Where Leakage Actually Hides

Context inflation nobody scoped, low-confidence runs re-tried without a fix, workflows nobody remembers enabling.

Leak

The tell: leakage rarely looks dramatic in a single run — it shows up as a slow upward trend across many runs of the same workflow.

Quality & Control Signals, and Adoption Tracking

Two different questions — “is it good?” and “is anyone using it?” — both worth tracking on purpose.

QUALITY & CONTROL SIGNALS

- Module 09's LLM-as-Judge scores, aggregated over time by workflow type
- Regression rate on brownfield changes — Module 06's discipline, measured
- Escalation rate — trending down could mean improving, or miscalibrating

ADOPTION TRACKING

- Who's actually running MCP workflows & studio automations, and how often
- Adoption by GBE — PA / IA / BA — not just an aggregate programme number
- Trend over the 30-day catch-up window from Module 01's free value-adds

Neither signal alone is enough: high adoption with falling quality is a warning, not a win — and vice versa.

Thresholds & Alerts: Turning Signals Into Action

A dashboard nobody checks is decoration — a threshold that fires is a operating rhythm.

Set a Baseline

Establish normal ranges for token spend, escalation rate & failure frequency per workflow.

Define the Threshold

State the number that should trigger a look — not “when it feels off.”

Alert Fires

A named person gets notified — the same discipline as Module 09's escalation path.

Operational Review

A regular rhythm — weekly or monthly — where thresholds themselves get revisited.

Common mistake: setting a threshold once and never revisiting it. Baselines shift as adoption grows — review the number, not just the alert.

Hands-On Lab: Ingest, Inspect, Compare

Four stages, real telemetry — finding out what your own engineering activity actually looks like.

Ingest

Load real or simulated engineering-agent run telemetry into Agent Prism.

Inspect

Step through traces; identify failures using Slide 8's six failure patterns.

Identify Leakage

Find where token/cost spend doesn't match Module 03's scoped-context discipline.

Compare Runs

Set a successful run against a poor-quality one; name exactly what differed.

Bring forward: the leakage and comparison findings from today become direct inputs to Module 11's token economics and ROI model.

Anti-Patterns: How Observability Gets Ignored

Five habits that turn a working monitoring lens into a dashboard nobody trusts.

DO THIS	NOT THAT
Set explicit thresholds and name who gets the alert	Build a dashboard and hope someone glances at it
Investigate a cost trend the first time it climbs	Wait until the token bill is a surprise to look
Treat a failure pattern as a signal to fix the workflow	Retry the same failing run and call it resolved
Track adoption by GBE, not just a single aggregate number	Report “engagement is up” without knowing where
Revisit baselines as adoption grows	Set a threshold once and never touch it again

Module 10 Outcomes & What's Next

Everything below should be true before we turn today's telemetry into an ROI model in Module 11.

- Can explain traces, replay & drift — and why each matters differently
- Can distinguish expected token spend from genuine cost leakage
- Can set a threshold and name who owns the resulting alert
- Compared a successful run against a poor-quality one and named the difference
- Know all six failure patterns and can identify them from real telemetry
- Understand quality/control signals and adoption tracking as two distinct questions
- Ingested real run data and inspected it for failures, hands-on
- Know the five anti-patterns that make a dashboard nobody trusts

NEXT — MODULE 11: ROI, TOKEN ECONOMICS AND DEPLOYMENT GOVERNANCE

1 hour · Cost per accepted output, saved effort, and governance decisions — built directly on today's traces, leakage findings & comparisons.

Questions & Discussion

Module 10 — Agent Prism for Monitoring, Observability and Enterprise Control